

HAGEN HAEUSSLER

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EDUCATION

University of California, Berkeley – Specialization

August 2025 – Present

- Fully funded (tuition fees and living costs) by a merit-based Scholarship from the German Academic Scholarship Foundation
- CS 169L: Software Engineering Team Project, ENGIN 183E: Technology Entrepreneurship, EECS 151: Intro. to Digital Design & ICs
- EECS 151LA: Application Specific Integrated Circuits Laboratory: SystemVerilog, Verdi, Genus, Innovus, VCS, HAMMER, SkyWater 130nm
- CS 188: Intro. to Artificial Intelligence, EECS 126: Probability and Random Processes, EECS 149: Embedded & Cyber-Physical Systems

Technical University of Munich (TUM) – B.Sc. Engineering Science

September 2023 – Present

- Broad, rigorous interdisciplinary engineering program (210 ECTS instead of typical 180 ECTS, selective admission)
- Awarded with the scholarships of the German National Academic Foundation (merit-based, top 0.5% German students) & the Federal German Scholarship "Deutschlandstipendium" (merit-based, top 1.2% of students in Germany)
- Grade: 2.4/5.0 (with 1.0 being the best and 5.0 being the worst); ~ top 20% of class
- Linear Algebra, Partial Differential Equations, Tensor Calculus, Physics, Chemistry, Mechanics, Fluid Dynamics, Thermodynamics, Control Theory, Electromagnetism, Bioengineering, Heat Transfer, Memory Hierarchy & Technology, Scheduling & Simulation, Reachability Analysis, Formal Verification, Pipelining, Latches & Flip-Flops, EDA, Static Timing Analysis
- Computer Architecture, Semiconductors, CAD, UML, CMOS Architecture, RISC Architecture, Combinational & Sequential Logic, FSMs
- RTL Design, Synthesis, FPGA, Physical Design, Timing Analysis, Data Structures, Databases, Search Algorithms, SQL, C, Java, Python

Parker School, Hawaii – High School Exchange Year

October 2020 – June 2021

- GPA 4.0 | AP Physics, Computer Science, 3D Design, Honors Mathematics | Awards: Head's List for "Outstanding Academic Performance"

Georg-Cantor-Gymnasium – High School

September 2014 – June 2023

- Abitur 1.0 (best possible grade, 2nd in class) | Focus: Mathematics & Natural Sciences (selective admission based on math & cognitive test)
- Awards: Membership in Germanys National Physics Academy (DPG), multiple State Math Olympiad & State Physics Olympiad awards

ENGINEERING & RESEARCH PROJECTS

Sim-to-Real Validation Research – UC Berkeley

August 2025 – December 2025

- Co-authored research paper on closing the Sim-to-Real gap using falsification-driven testing on Pololu 3pi+ robot
- Developed methodology to identify where robot simulations fail in reality – critical for safe deployment of autonomous systems
- Paper: https://www.hagenhaeussler.com/Sim2Real_Research_Project_UC_Berkeley_2025.pdf

Embedded Cyber-Physical Systems Lab – UC Berkeley

August 2025 – Present

- Designing embedded systems using Lingua Franca with C for timing, concurrency, PID-controlling and state-machine modeling
- Implementing low-level software interfacing with sensors/actuators on Pololu 3pi+ 2040 robot and Raspberry Pi Pico

Hardware Engineering Projects – Independent Work

- Solar-Powered Climate Control: Designed photovoltaic-powered cooling device integrating electrical systems & thermodynamics
- Ballistic Optimization (Trebuchet): Applied physics principles and iterative testing to maximize performance & win school competition

WORK EXPERIENCE

KSB SE & Co. KGaA – Industrial Mechanics Internship

October – November 2018

- Manufactured high-precision machine components using industrial lathes and milling machines

SONOTEC GmbH – Internship & Shadowing Program

May 2019

- Built non-contact ultrasonic sensor for industrial tank quality monitoring

Porsche (Leipzig) – Servicing and Sensor Calibration

July 2022

- Reconfigured arc-welding monitoring systems for HKS Prozesstechnik GmbH

PERSONAL & SKILLS

- **Judo Black Belt (1. Dan):** Central German Judo Championships U21: 1st Place
- **CANTOR.open "Strongest Student":** 1st Place in city-wide strength competition
- **Tools:** SolidWorks, MATLAB, VS Code, DaVinci Resolve, 3D printing, laser cutting, Arduino, 3pi+ 2040 Robot, soldering, Inkscape
- **Leadership:** 8x Class President, 2x School President, founded and scaled school's first Christmas Market and YouTube channel
- **Munich Music Association (AGVM):** Manage social media reaching 3000+ followers (Insta: @agvmuenchen, Facebook: Socialmedia Agv)